Total Body Water Deficit *Prodigy Anesthesia*

Total body water (TBW) deficit is a calculation used to determine the number of liters of water that should be administered to a patient with hypernatremia. Note that the water deficit, once calculated, should be administered over a period of 24 to 48 hours to prevent the development of cerebral edema and/or seizures from too rapid a correction of the sodium level. When correcting hypernatremia, one should not lower the level more than 1-2 mEq/L per hour.

The formula for calculating total body water deficit is as follows:

TBW deficit = (0.6 X body weight in kg) X [(Serum sodium level - 140)/140]
*note that 140 is used in this calculation because it is the middle of the normal sodium range

EXAMPLE

Calculate the TBW deficit for an 80 kg patient with a serum sodium of 154 mEq/L.

TBW deficit = (0.6 X 80 kg) X [(154-140)/140]

TBW deficit = (48) X [14/140]

TBW deficit = 48 X 0.1

TBW deficit = 4.8 Liters

PRACTICE PROBLEMS

1. Calculate the total body water deficit for a 220 pound patient with a serum sodium of 155 mEq/L.
2. Calculate the total body water deficit for a 98 kg patient with a serum sodium of 158 mEq/L.
3. Calculate the total body water deficit for a 119 pound patient with a serum sodium of 152 mEq/L.
4. Calculate the total body water deficit for 78 kg patient with a serum sodium of 156 mEq/L.
5. Calculate the total body water deficit for a 289 pound patient with a serum sodium of 158 mEq/L.

ANSWERS

1. TBW deficit = (0.6 X 220/2.2) X [(155-140)/140]
   TBW deficit = (60) X (15/140)
   TBW deficit = 6.43 L

2. TBW deficit = (0.6 X 98) X [(158-140)/140]
   TBW deficit = (58.8) X (18/140)
   TBW deficit = 7.56 L

3. TBW deficit = (0.6 X 119/2.2) X [(152-140)/140]
   TBW deficit = (32.45) X (12/140)
   TBW deficit = 2.78 L

4. TBW deficit = (0.6 X 78) X [(156-140)/140]
   TBW deficit = (46.8) X (16/140)
   TBW deficit = 5.35 L

5. TBW deficit = (0.6 X 289/2.2) X [(158-140)/140]
   TBW deficit = (78.8) X (18/140)
   TBW deficit = 10.1 L

Barash PG, Cullen BF, Stoelting RK. Clinical Anesthesia. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 193.